Climate Change: Causes and Effects

Knowledge document 031 of 100

Introduction to Climate Change

Climate change refers to long-term shifts in temperatures and weather patterns across the planet. Natural factors such as volcanic eruptions and variations in the Sun's energy have caused climate changes throughout Earth's history, but since the 1800s the main driver has been human activity, primarily the burning of fossil fuels such as coal, oil and natural gas. The Intergovernmental Panel on Climate Change (IPCC), set up in 1988 by the World Meteorological Organization and the United Nations Environment Programme, has concluded that it is unequivocal that human influence has warmed the atmosphere, ocean and land. Global average temperature is now about 1.1 to 1.2 degrees Celsius above the pre-industrial level (1850 to 1900), and 2024 was the warmest year on record, the first calendar year to exceed 1.5 degrees Celsius above the pre-industrial average according to several agencies.

What Is the Greenhouse Effect?

The greenhouse effect is a natural process that warms the Earth. Sunlight passes through the atmosphere and heats the Earth's surface. The surface then radiates heat back as infrared radiation. Certain gases in the atmosphere, called greenhouse gases, absorb part of this outgoing heat and send some of it back toward the surface, like the glass roof of a greenhouse that traps warmth. Without the natural greenhouse effect, the average temperature of the Earth would be about minus 18 degrees Celsius instead of the present comfortable average of about 15 degrees Celsius, and life as we know it could not exist. The problem of climate change is the enhanced greenhouse effect: human activities have added huge extra amounts of greenhouse gases, so more heat is trapped and the planet warms. The effect was first described by Joseph Fourier in the 1820s, and Svante Arrhenius calculated in 1896 that doubling carbon dioxide would warm the planet.

Three Greenhouse Gases

Three major greenhouse gases are carbon dioxide, methane and nitrous oxide. Carbon dioxide (CO2) is the most important, released by burning coal, oil and gas, by cement production and by deforestation. Its concentration in the atmosphere has risen from about 280 parts per million before the Industrial Revolution to over 420 parts per million today, the highest in at least 800,000 years. Methane (CH4) comes from cattle, rice paddies, landfills, coal mines and natural gas leaks; it is more than 25 times as powerful as carbon dioxide at trapping heat over 100 years, but it stays in the air for only about a decade. Nitrous oxide (N2O) comes from fertilisers, soils and industry and is about 270 times as potent as CO2 per molecule. Others include water vapour, which amplifies warming but responds to temperature, ozone, and fluorinated gases such as CFCs, HFCs and sulphur hexafluoride, which are very powerful.

Main Causes

The burning of fossil fuels for electricity, heat, transport and industry is responsible for roughly three-quarters of greenhouse gas emissions. Deforestation removes trees that absorb carbon dioxide. Agriculture, including livestock and fertiliser use, emits methane and nitrous oxide. Industrial processes, such as steel and cement production, and waste in landfills also contribute. The largest emitters today are China, the United States and India, although per-person emissions in India are far below the world average.

Effects of Climate Change

Rising temperatures cause more frequent and severe heatwaves. Glaciers and ice sheets are melting, and global mean sea level has risen about 20 centimetres since 1900 and is now rising by over 4 millimetres each year, threatening low-lying coasts and islands. Oceans absorb carbon dioxide and are becoming more acidic, damaging corals and shellfish, and warm seas cause coral bleaching. Rainfall patterns are changing, with more intense downpours, floods and cyclones in some areas and longer droughts in others. Crop yields and food security are at risk, water shortages worsen, diseases spread to new areas, and many species face extinction. The poorest communities are the most vulnerable.

Climate Change and India

India is highly vulnerable because of its long coastline, dependence on the monsoon, large farming population and Himalayan glaciers that feed major rivers. The 2015 Chennai floods and the heavy rain of Cyclone Michaung in December 2023 showed the growing risk of extreme rainfall on the Tamil Nadu coast. Heatwaves in north India, glacial lake floods in the Himalaya and rising sea levels along the coast are also concerns. India launched the National Action Plan on Climate Change in 2008, and at the COP26 summit in Glasgow in November 2021 Prime Minister Narendra Modi announced five commitments, called Panchamrit: reaching 500 gigawatts of non-fossil electricity capacity by 2030, meeting 50 percent of energy needs from renewables by 2030, reducing projected carbon emissions by one billion tonnes by 2030, cutting the carbon intensity of the economy by 45 percent by 2030, and achieving net-zero emissions by 2070. Tamil Nadu has set up a Climate Change Mission and promotes wind and solar power.

The Paris Agreement

The Paris Agreement was adopted on 12 December 2015 at the COP21 conference in Paris and entered into force on 4 November 2016. Its central goal is to limit global warming to well below 2 degrees Celsius above pre-industrial levels, and to pursue efforts to limit the increase to 1.5 degrees Celsius. Each country submits its own national climate plan called a Nationally Determined Contribution (NDC), updated every five years. Earlier, the Kyoto Protocol of 1997 had set binding targets for developed countries. The goal is to reach net-zero emissions in the second half of the century, which means balancing the greenhouse gases released with those removed.

Solutions: Mitigation and Adaptation

Mitigation means reducing emissions through renewable energy, energy efficiency, electric vehicles, protecting forests, planting trees and cutting waste. Adaptation means adjusting to the changes, for example building flood defences, growing drought-resistant crops and improving early warning systems. Individuals can help by saving electricity, using public transport, reducing food and plastic waste and planting trees.

Challenges and Future

Challenges include dependence on fossil fuels, the cost of the transition, unequal responsibility between rich and poor nations, and slow international action. Technologies such as green hydrogen, batteries, carbon capture and nuclear power may help. The decisions taken this decade will determine how much warming occurs.

Quick Facts

- The greenhouse effect is the trapping of heat by gases in the atmosphere; it keeps Earth at about 15 degrees Celsius.
- Three greenhouse gases are carbon dioxide, methane and nitrous oxide.
- The Paris Agreement (2015) aims to limit warming to well below 2 degrees Celsius and pursue 1.5 degrees Celsius.
- India has pledged net-zero emissions by 2070.
- Atmospheric carbon dioxide has risen from about 280 ppm to over 420 ppm.